User Manual

Students will be provided an easy-to-use GUI web to operate robots. It's intended to reduce the burden of ROS coding and allow students to focus on practicing scheduling. This document explains the user interface and basic operation methods for controlling two TurtleBot robots in the final project.

1. Power On

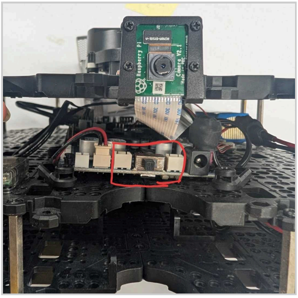

Fig. 1: Power switch on the front of the robot

The power switch is on the front of the robot.

1. Place the robot at its designated start station.
2. Flip **the switch** to the ON position.
3. Wait **2–3 minutes** for the robot to boot and connect.
4. The robot appears automatically in the web dashboard.

Note: If the robot is turned on at a different station or angle from its physical location, perform **Relocalize** (Section 4.3) before issuing any move commands.

Important: Always **power off** the robot (flip the switch to OFF) when you finish using it.

2. Battery Replacement

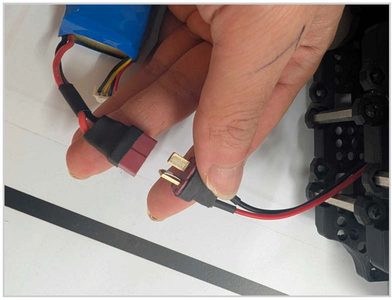

Fig. 2: Li-Po battery

When the Li-Po battery is depleted, the robot halts and emits a **beep** sound.

1. Flip the power switch OFF.
2. Remove the battery.
3. Disconnect the battery connector.
4. Connect a charged spare and place it back.
5. Power on; the robot reconnects after ~2–3 min.

Note: Charged spare batteries are on the charging station beside the testbed. Return depleted batteries to the charger immediately.

3. Laptop

The ROS server and the Web UI run on the laptop automatically. You do **not** need to start or stop anything by hand.

- You can safely **power off**, **sleep**, **suspend**, or **reboot** the laptop at any time.
- **Recommended: keep the laptop on 24/7.** You don't have to turn it off actually.
- If the laptop reboots, wait about **1 minute**, then open **http://localhost:5173** in the browser and refresh the page.
- Login password is **1234**.

📞 **Contacts**

TA	**Wonseok Jang**	cotidie@kaist.ac.kr	*ROS & System*
TA	**Mingyu Suh**	mg0208@kaist.ac.kr	*ROS & System*
TA	**Sangjin Ban**	sahngjin.ban@kaist.ac.kr	*Scheduling & Simulation*

4. Web UI Operations

Open **http://localhost:5173** on the provided LG Gram laptop. The left panel shows the graph map with live robot positions; the right panel shows the selected robot's status and controls.

4.1 Move

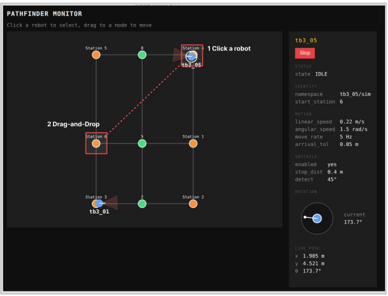

Fig. 3: Click a robot, then drag it to the target node

1. **Click** a robot icon to select it (highlight box appears).
2. **Drag** the robot to a destination graph node and release.
3. The server plans a Dijkstra shortest path and dispatches it.

4.2 Stop

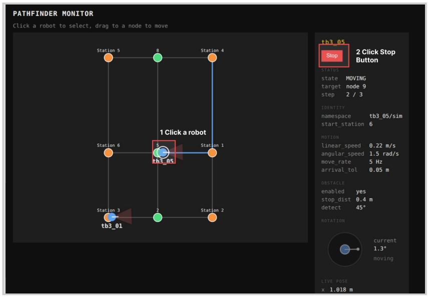

Fig. 4: Click a robot, then press the Stop button

1. **Click** a robot on the map to select it.
2. Click the red **Stop** button in the right panel.
3. The robot halts immediately; state returns to `IDLE`.

4.3 Relocalize

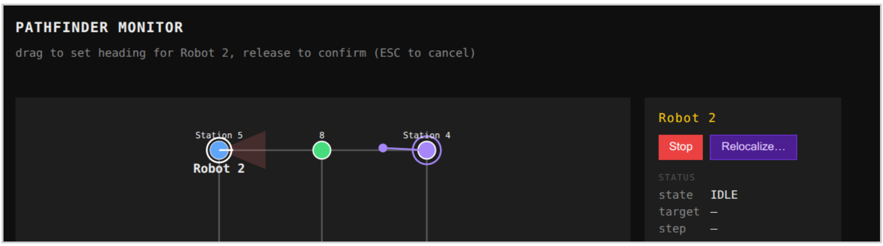

Fig. 5: Drag to set heading after a manual placement

Use this whenever the robot's real location or heading no longer matches what is shown on the map, for example after physically moving the robot or powering on at a different station or angle.

1. **Click** a robot to select it.
2. Click the purple **Relocalize…** button.
3. **Drag** on the map to set the heading; release to confirm. Press **ESC** to cancel.
4. The pose estimate updates; no motion is sent to the robot.

5. Quick Reference

Action	How	Result
Power on	Flip both switches ON	Robot appears in dashboard after ~2–3 min
Move	Click robot → drag to node	Dijkstra path dispatched; state = `MOVING`
Stop	Click robot → press Stop	Immediate halt; state = `IDLE`
Relocalize	Click robot → Relocalize… → drag heading	Pose updated; no motion
Replace battery	Power off → swap battery → power on	Robot reconnects after ~2–3 min